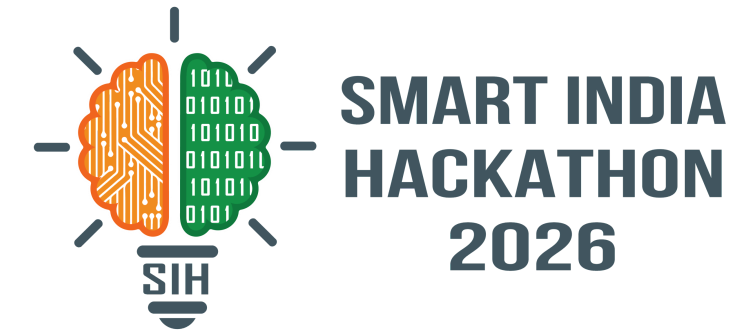
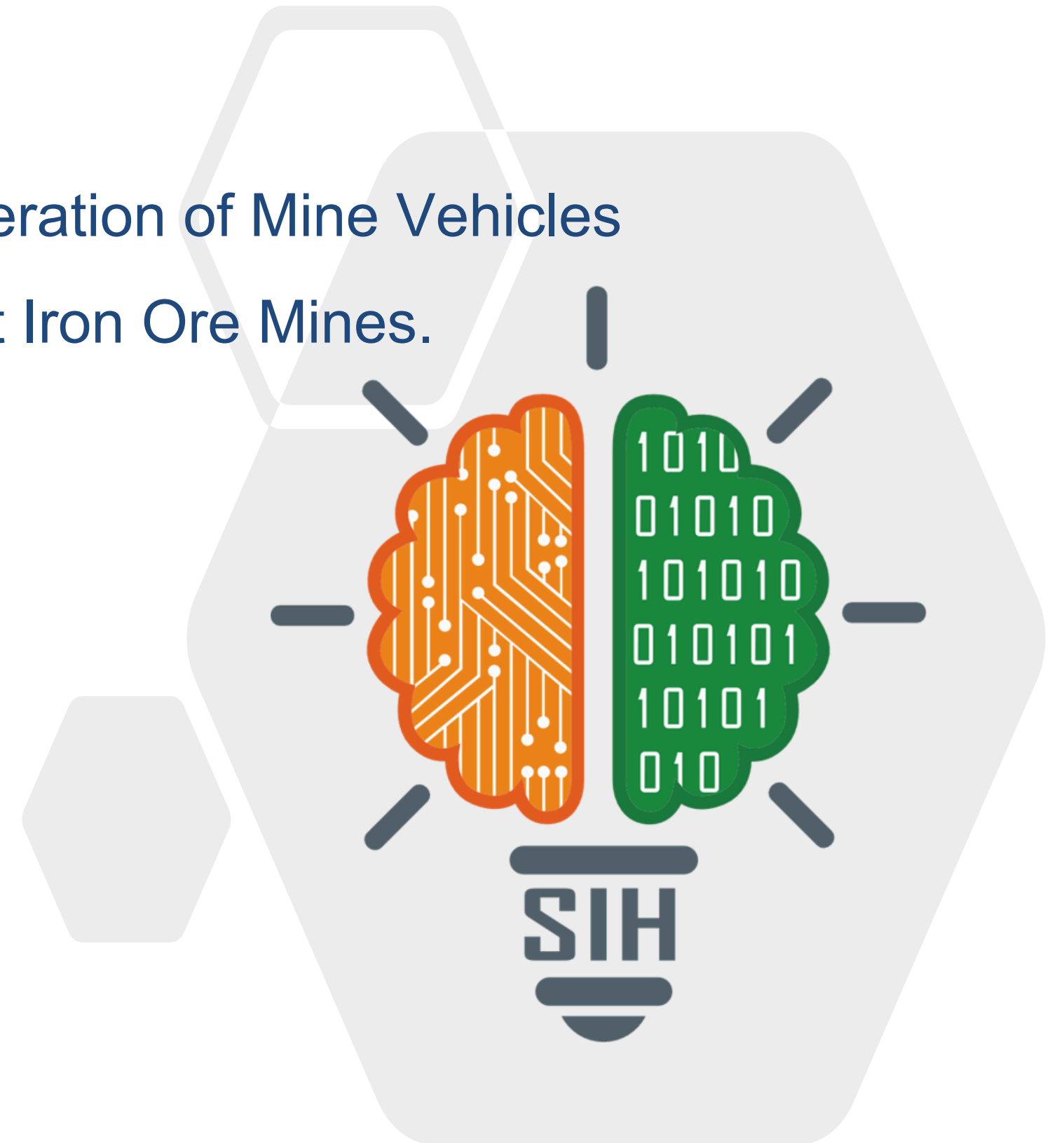


SMART INDIA HACKATHON 2026



- **Problem Statement ID-** SIH26007
- **Problem Statement Title-** Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.
- **Theme-** Smart Automation
- **PS Category-** Hardware
- **Team ID-**
- **Team Name-** Team Innov8



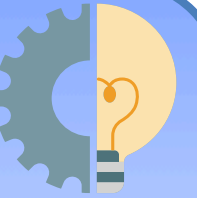
Problem



- At Bailadila, dense monsoon fog can obscure dumper visibility and increase collision risk. current aids do little.
- Safety forces slow-downs or full halts, increasing cycle times and delaying ore evacuation.
- Result: a major production and fleet output loss every foggy day.
- There is currently no smart technology that allows vehicles to continue travelling in fog. it's just slow-or-stop.



Our Solution



- Cameras + AI help trucks see through fog and dust.
- AI predicts fog's arrival before it develops.
- Trucks can still communicate where they are to each other in zero visibility.
- This system can automatically reduce the speed of trucks depending on fog levels.
- Control room dashboard shows live truck positions and reroutes them as necessary.

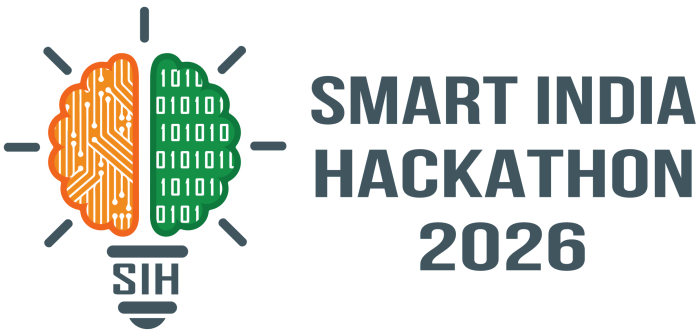
Why Different



- **Fog-blind detection:-** Multy sensors + dust-calibrated AI vision finds obstacles, even when standard cameras can't see through the haze.
- **Constantly updating severity:-** 0-5 Fog Severity Index automatically updates based on current visibilty and dust conditions.
- **AI calculated speed:-** Automatically calculates a safe speed based on DGMS's braking distance rule, rather than requiring a full stop/restart.
- **No-visibility coordination:-** GPS-meshed system lets all vehicles know where the others are, even when they can't see them.
- **Live geo-map:-** Operators can view current vehicles on a digital-twin map, and reroute as needed based on developing fogs.



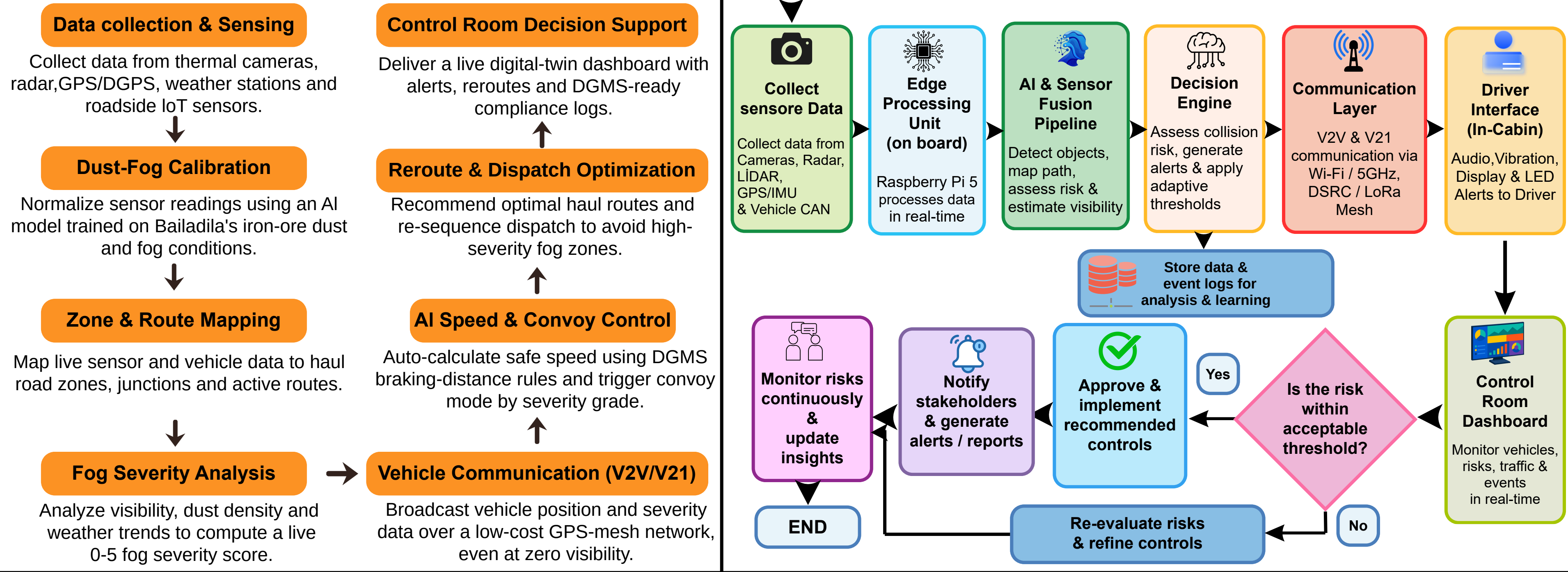
TECHNICAL APPROACH



Team
Innov8

Methodology

Flow chart



ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY

FogGuard can be implemented using proven, low-cost technologies: thermal cameras, GPS, LoRa mesh and lightweight AI - so it can be installed on existing dumpers without requiring new vehicles.

2. ECONOMICAL FEASIBILITY

The retrofit kit costs a fraction of imported collision-avoidance systems. Using open-source software and cheap sensors, mines recover costs through fewer accidents and less downtime.

3. SOCIAL FEASIBILITY

FogGuard creates trust inside the mine by giving drivers, pedestrians and light-vehicle operators real-time visibility warnings. This reduces the fear associated with monsoon weather.

4. LEGAL & COMPLIANCE FEASIBILITY

FogGuard is based on DGMS's own visibility standards 3x braking distance 30m corner visibility and automatically generates DGMS compliant logs for safety audits and reporting.

5. OPERATIONAL FEASIBILITY

Automated detection of fog severity, real-time alerts and dashboard rerouting eliminate guesswork for drivers and control room personnel, enabling smooth haulage

6. SECURITY FEASIBILITY

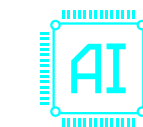
Using vehicle-to-vehicle data and the control room dashboard requires secure communication, role-based access and audit logging to keep safety data safe from unauthorized access, thereby ensuring that the information is not tampered with.

POTENTIAL CHALLENGES AND RISKS

SENSOR RELIABILITY IN FOG AND DUST



Iron-ore dust mixed with fog can confuse the camera, leading to false alerts or missed detections



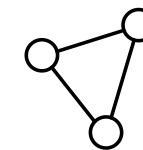
DUST-CALIBRATED AI TRAINING

Train and refine the vision model on actual site dust / fog to reduce false positives

CONNECTIVITY AND COVERAGE



Remote haul roads can have poor network coverage, introducing delays in the transmission of live vehicle data.



OFFLINE-FIRST MESH NETWORK

Use LoRa mesh + on-device processing so the system doesn't rely on constant internet connection



OPERATOR TRUST & ALARM FATIGUE

Drivers can become conditioned to ignore or turn off alarms when they feel like they are coming up too often, not specific enough or otherwise unhelpful.



EXPLAINABLE, GRADED ALERTS

Provide transparency about the reason for each alert and build driver trust by using progressive escalation



FLEET-WIDE ROLLOUT COST

The cost of outfitting an entire fleet of dump trucks at once could be prohibitive for some mid-sized mining operations.



PHASED, PRIORITY ROLLOUT

First install on highest-incident haul routes to prove value before rolling out to entire fleet

IMPACT AND BENEFITS

IMPACTS

Fewer Fog Accidents

Minimizes collision risk during monsoon fog by enabling dumpers to see, communicate and slow down automatically.



Continuous Ore Evacuation

Ensures continuity of haulage even in fog, rather than shutting it down completely, thus protecting daily production targets.



DGMS Compliance Ready

Auto-logs every fog alert and every change in speed in a format compliant with the DGMS safety regulations.



Lower Retrofit Cost

Provides fog safety at a fraction of the cost of imported collision avoidance systems.



Live Fleet Visibility

Provides situational awareness of all vehicles in terms of their location and the severity of fog around them, in real-time to the control room.



BENEFITS

Data-Driven Dispatch

Lets dispatchers reroute trucks using live fog and traffic data, not guesswork.



Fleet Downtime

Cuts idle time caused by full-stop fog halts through graded, automatic speed control.



AI-Powered Fog Insight

Uses AI/ML to detect, grade and predict fog severity for smarter, faster decisions.



Optimized Speed Control

Balances safety and speed automatically, maximizing haulage within safe limits.



Safety-Production Balance

Bridges the gap between strict safety needs and the mine's daily production goals.



THE VALUE TRANSFORMATION



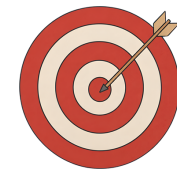
SENSOR DATA

Thermal, radar, GPS and weather signals, collected live



FOG SEVERITY INSIGHT

AI turns raw signals into a live 0-5 severity score



SMART SPEED CONTROL

Auto-sets safe speed and convoy mode per DGMS rules



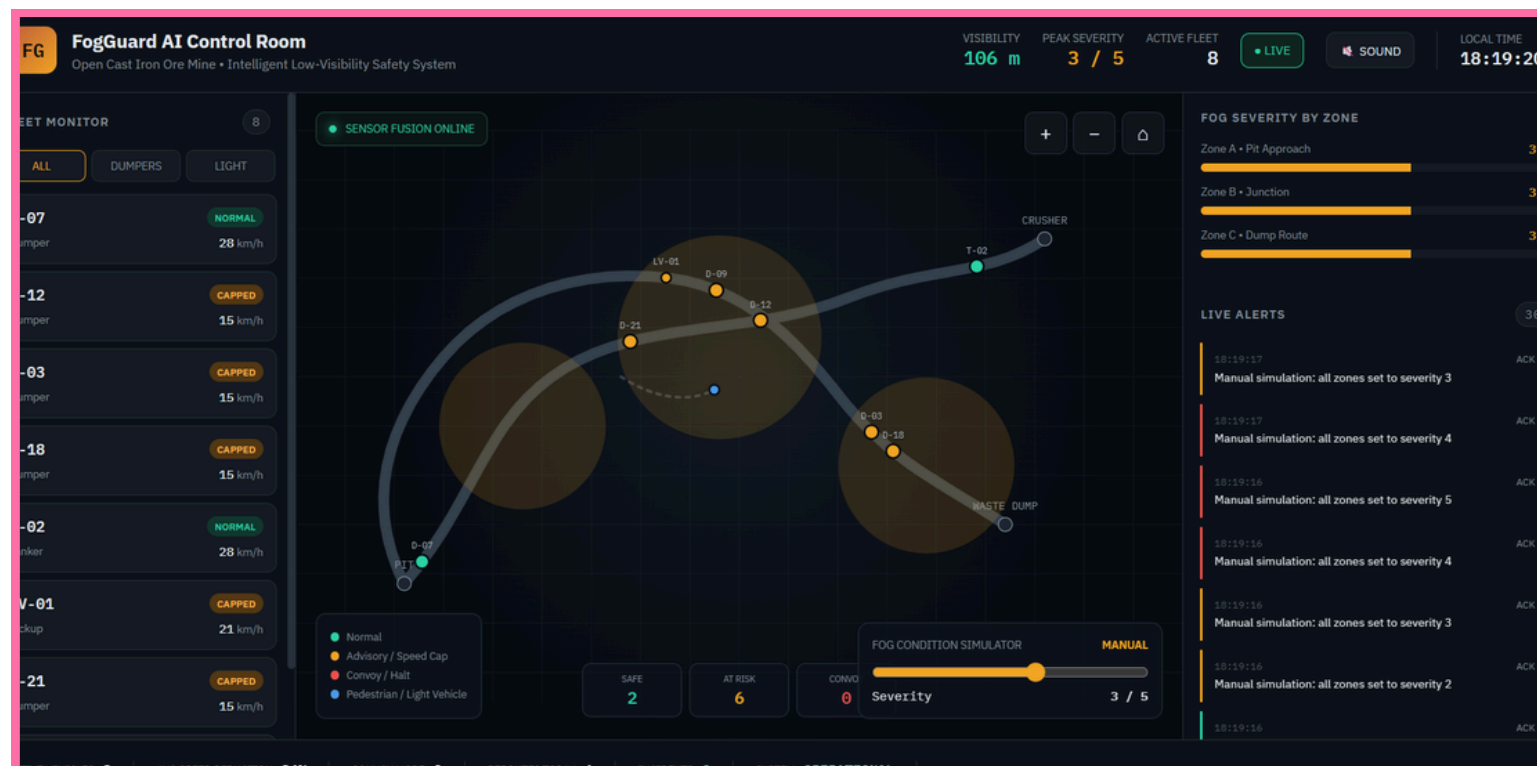
SAFE, CONTINUOUS HAUL

Fewer accidents. Less downtime. Reliable evacuation.



From raw fog data to safe, continuous mining operations.

Controle room dashboard



Vehical Interface dashboard



RESEARCH AND REFERENCES

Academic validation

- Chaulya, S. et al. (2025). "Smart driving assistance system for mining operations in foggy environments."
- Qian, M., Zhao, K., Li, B., Gong, H., & Seneviratne, A. (2022). "Survey of Collision Avoidance Systems for Underground Mines: Sensing Protocols."
- Accident analysis of mobile mine machinery in Indian opencast coal mines. International Journal of Injury Control and Safety Promotion. PMID 23324038.

Regulatory and statutory basis

- DGMS (Tech) Circular No. 06 of 2020 — proximity warning device requirements for HEMM.
- DGMS (Tech) Circular No. 09 of 2008 — maximum permissible speed in mines.
- Coal Mines Regulations 2017 / Metalliferous Mines Regulations 1961 / Mines Act 1952

Technical documentation and datasheets

- Texas Instruments AWR-series 77GHz FMCW mmWave radar — datasheet & range/Doppler processing guide
- Ultralytics YOLO documentation
- Espressif ESP-NOW protocol
- Raspberry Pi 5 documentation
- LoRa long-range sub-GHz link budget notes